Acommune Tech Pvt Ltd

presents

Complete Professional Course

Fusion 360

From Concept to Manufacturing — A Complete Learning Journey

Course Code	ACT-FUS-2025
Duration	80 Hours | 8 Modules | 40 Sessions
Level	Beginner to Advanced
Format	Instructor-Led + Hands-On Projects

Certification	acommune Tech Pvt Ltd Certificate of Completion

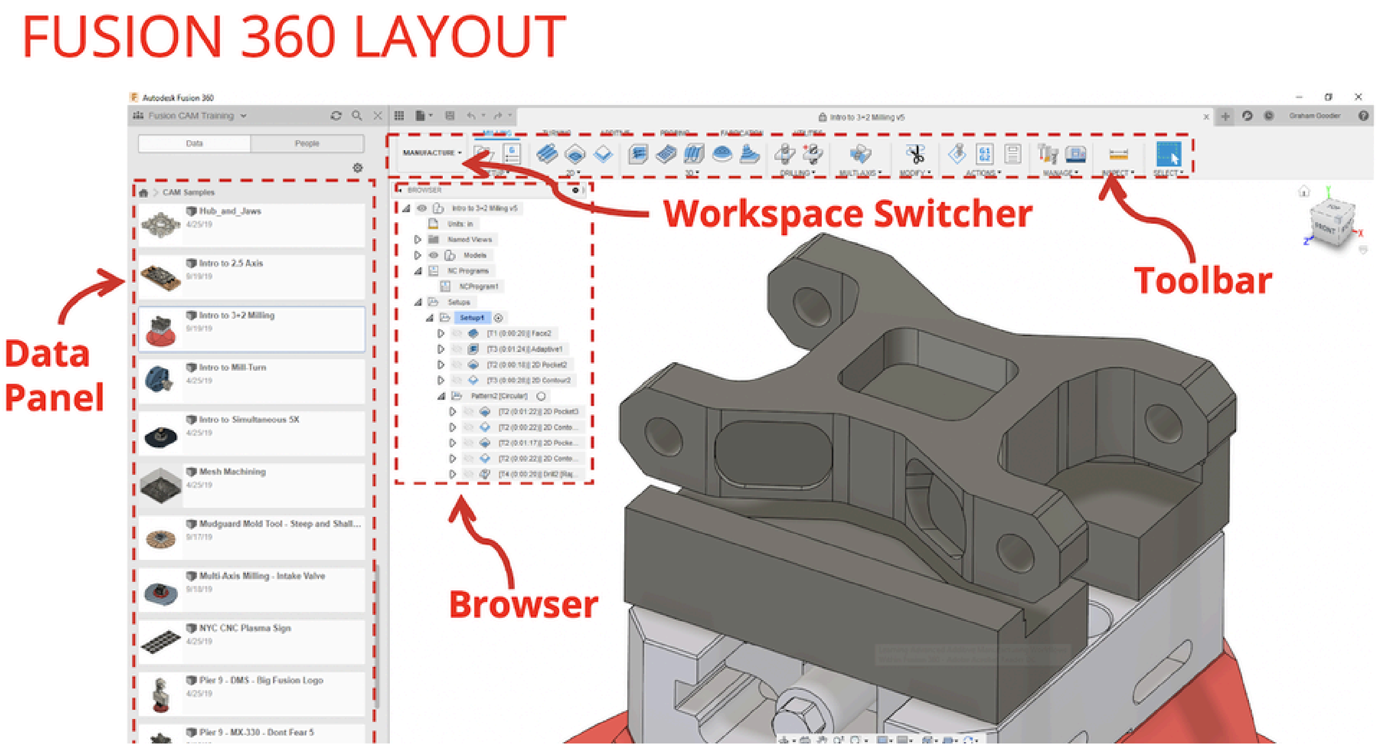
Issued By	acommune Tech Pvt Ltd | training@acommune.tech

This course is structured across 8 comprehensive modules spanning 20 sections, designed to take learners from beginner fundamentals to advanced manufacturing workflows using Autodesk Fusion.

About This Course

Autodesk Fusion is the industry-leading cloud-based product development platform integrating 3D CAD, CAM, CAE, and PCB design into a single unified environment. This course, designed and delivered by acommune Tech Pvt Ltd, equips learners with end-to-end skills required for modern engineering and product development roles.

Through structured modules, real-world projects, and guided practice sessions, participants will develop proficiency in designing, simulating, manufacturing, and presenting engineering products using Autodesk Fusion.



Module 1 Introduction to Autodesk Fusion

This foundational module introduces learners to the Autodesk Fusion environment, its cloud-based infrastructure, and core workflows. Participants learn to navigate the interface, manage cloud documents, and understand the integrated design-to-manufacturing philosophy behind Fusion.

1.1 Overview of Autodesk Fusion

Autodesk Fusion (formerly Fusion 360) is a cloud-based platform that bridges the gap between design, engineering, and manufacturing. This section covers its evolution, key differentiators, and industry applications.

✓ History & evolution from Fusion 360	✓ Subscription plans & licensing overview
✓ Cloud vs. desktop CAD comparison	✓ System requirements & installation
✓ Industry use cases (Aerospace, Auto, Consumer)	✓ Autodesk account setup

1.2 Interface & Workspace Navigation

Mastering the Fusion interface is the first step to productivity. Learners explore panels, toolbars, the data panel, and keyboard shortcuts essential to efficient workflows.

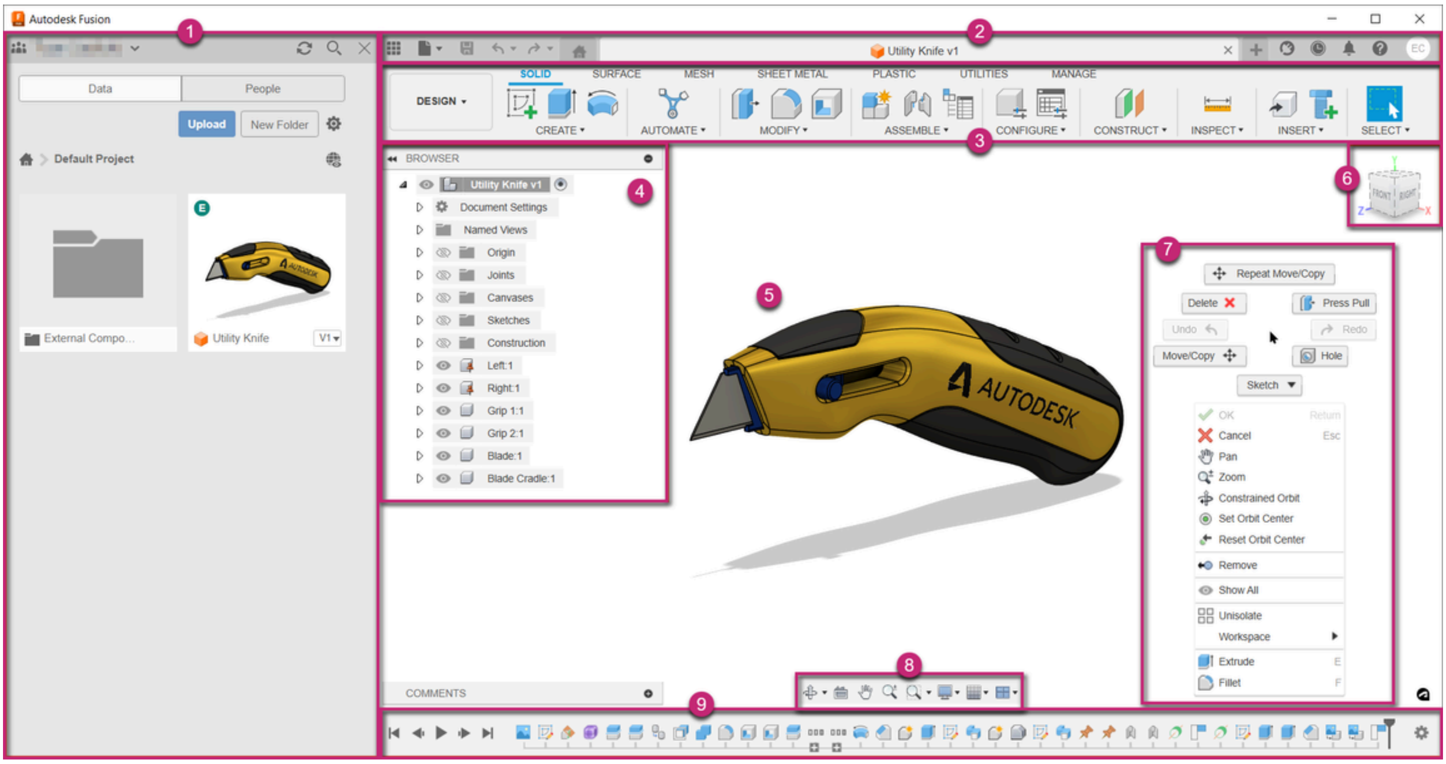
✓ Toolbar & ribbon customization	✓ View cube & orbit controls
✓ Data panel & cloud file management	✓ Component browser & timeline
✓ Design, Manufacture, Simulation workspaces	✓ Keyboard shortcuts & preferences

1.3 Basic Workflow & Project Setup

Understanding the parametric and history-based design approach in Fusion. Participants create their first design project and manage version history.

✓ Parametric vs. direct modeling concepts	✓ Importing/exporting file formats (STEP, IGES, STL)
✓ Creating & saving design projects	✓ Units & document settings
✓ Cloud version control & branching	✓ Collaboration & sharing designs

Module Duration: 8 Hours | Sessions: 4 | Lab Exercises: 3



Module 2 Sketching & 2D Drawing Fundamentals

2D sketching is the foundation of all 3D modeling in Fusion. This module teaches learners to create precise, fully-constrained sketches using geometric and dimensional constraints, forming the basis of parametric solid models.

2.1 Sketch Environment & Tools

✓ Entering & exiting sketch mode	✓ Spline & conic curves
✓ Sketch planes & coordinate system	✓ Polygon & slot tools
✓ Line, arc, circle, rectangle tools	✓ Construction geometry usage

2.2 Constraints & Dimensions

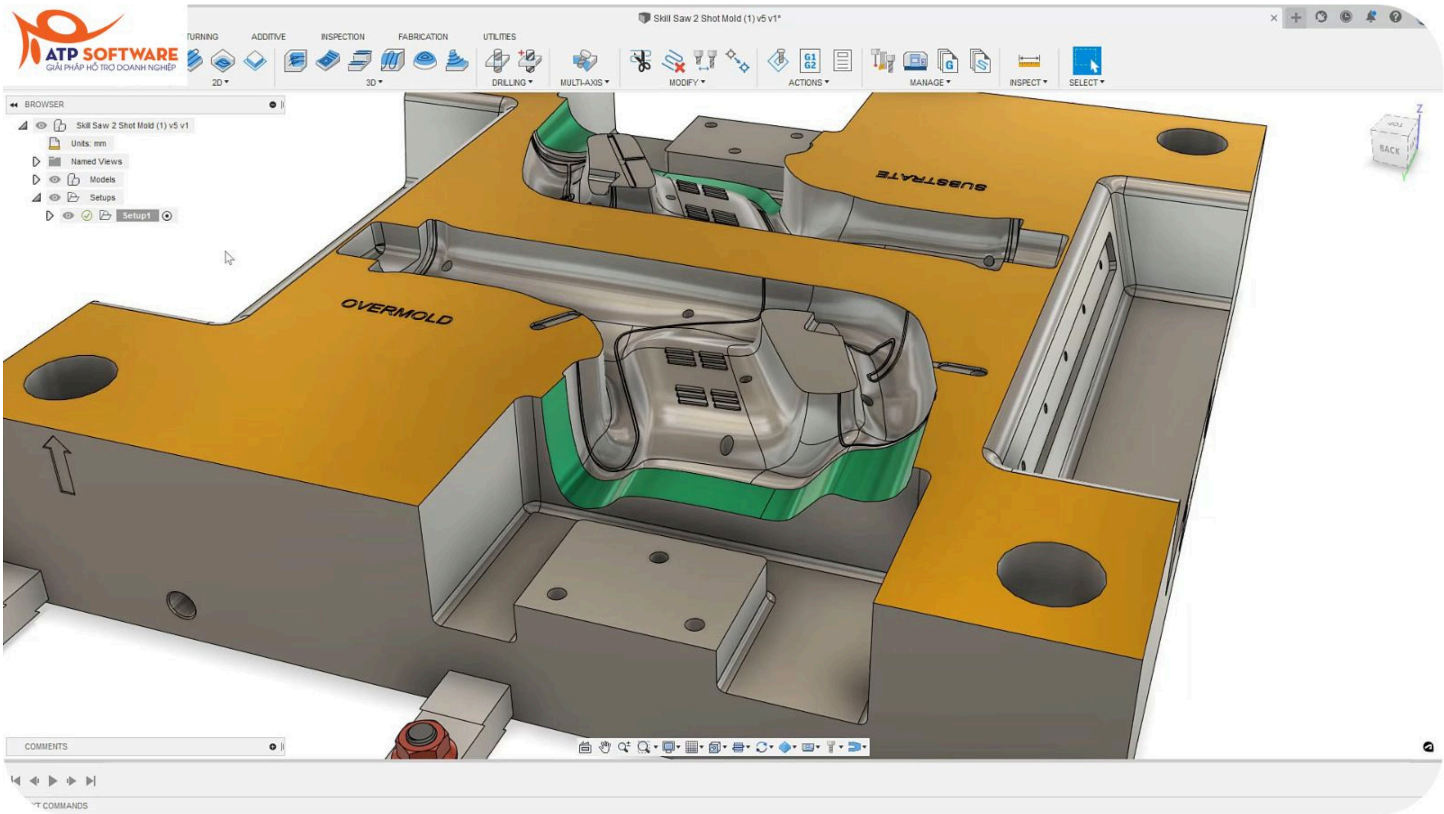
Fully constrained sketches ensure design intent is maintained throughout the parametric design process. This section covers all geometric and dimensional constraints.

✓ Geometric constraints (parallel, perpendicular)	✓ Linear, angular & radial dimensions
✓ Coincident, tangent, concentric constraints	✓ Driven vs. driving dimensions
✓ Horizontal & vertical constraints	✓ Over-constrained sketch resolution

2.3 Advanced Sketch Techniques

✓ Mirror & pattern in sketch	✓ Sketch from imported DXF files
✓ Offset & trim/extend tools	✓ Multi-body sketch profiles
✓ Project & include geometry	✓ Sketch best practices for 3D modeling

Module Duration: 10 Hours | Sessions: 5 | Lab Exercises: 4



Module 3 3D Solid & Surface Modeling

This core module covers the creation of 3D solid and surface models using Fusion's powerful parametric modeling tools. Learners apply sketch profiles to generate complex 3D geometry using extrude, revolve, sweep, loft, and advanced surface operations.

3.1 Solid Modeling Tools

✓ Extrude — single & multi-profile	✓ Hole, thread & chamfer/fillet tools
✓ Revolve & Sweep operations	✓ Shell & draft operations
✓ Loft with guide rails & profiles	✓ Boolean operations (Join, Cut, Intersect)

3.2 Feature-Based Design

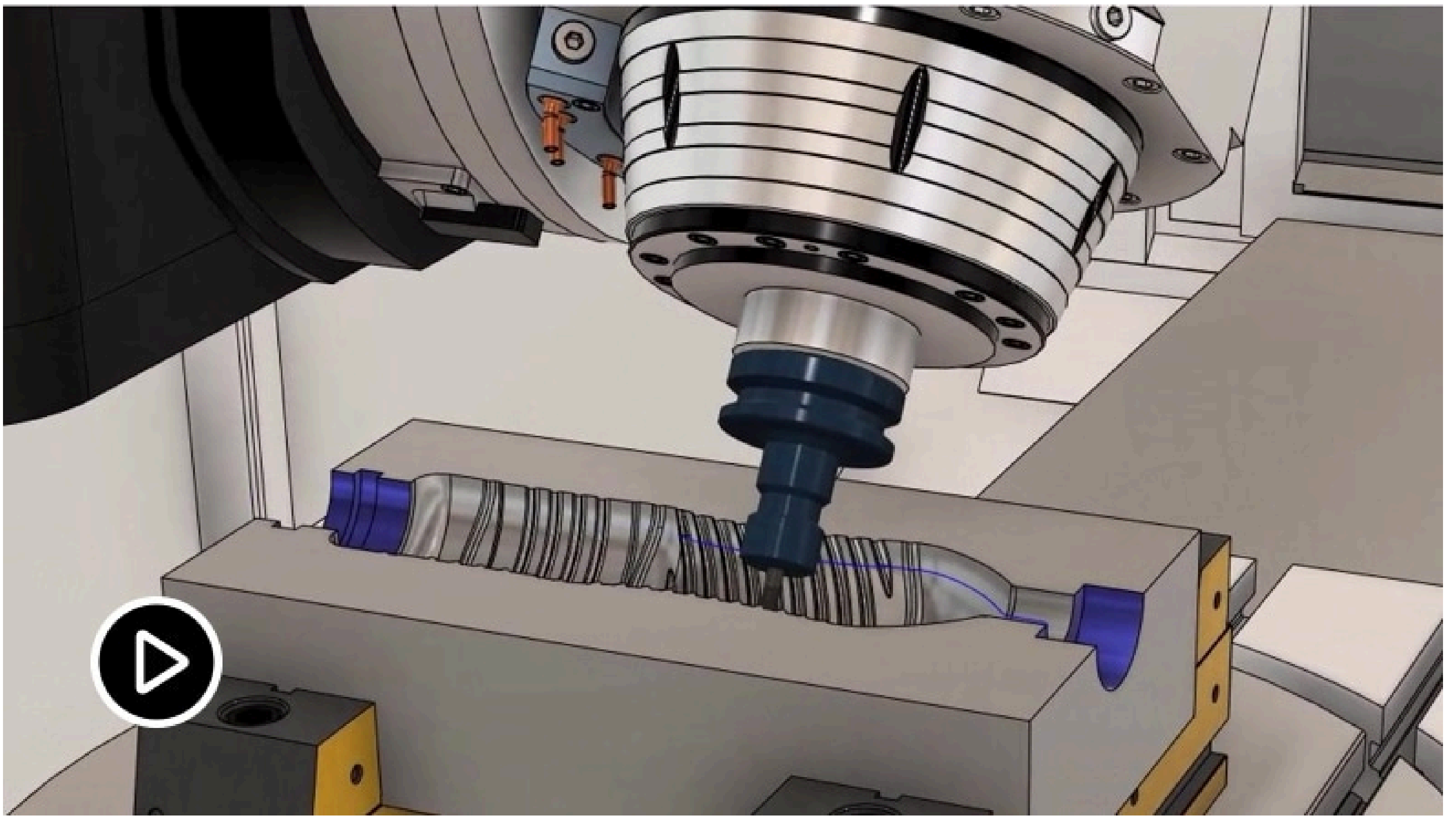
Feature-based modeling enables design modifications through the parametric timeline. Learners understand history management and feature suppression for efficient design iteration.

✓ Timeline management & reordering	✓ Rib, web & grill features
✓ Feature suppression & rollback	✓ Rule fillet & chamfer strategies
✓ Mirroring & pattern features	✓ Multi-body part design

3.3 Surface Modeling (Class A Surfaces)

✓ Surface vs. solid modeling approach	✓ Thicken & offset surface tools
✓ Patch, ruled & loft surfaces	✓ T-spline/Sculpt workspace intro
✓ Trim, untrim & stitch surfaces	✓ Converting surfaces to solids

Module Duration: 12 Hours | Sessions: 6 | Lab Exercises: 5



Module 4 Assembly Design & Motion Studies

Real-world products consist of multiple components assembled together. This module teaches learners to build complex assemblies in Fusion using joints, contacts, and motion studies to validate mechanical function before physical prototyping.

4.1 Assembly Fundamentals

✓ Bottom-up vs. top-down assembly	✓ As-built vs. parametric joints
✓ Creating & inserting components	✓ Rigid, revolute & slider joints
✓ Grounding & anchoring components	✓ Cylindrical, ball & planar joints

4.2 Assembly Constraints & Contacts

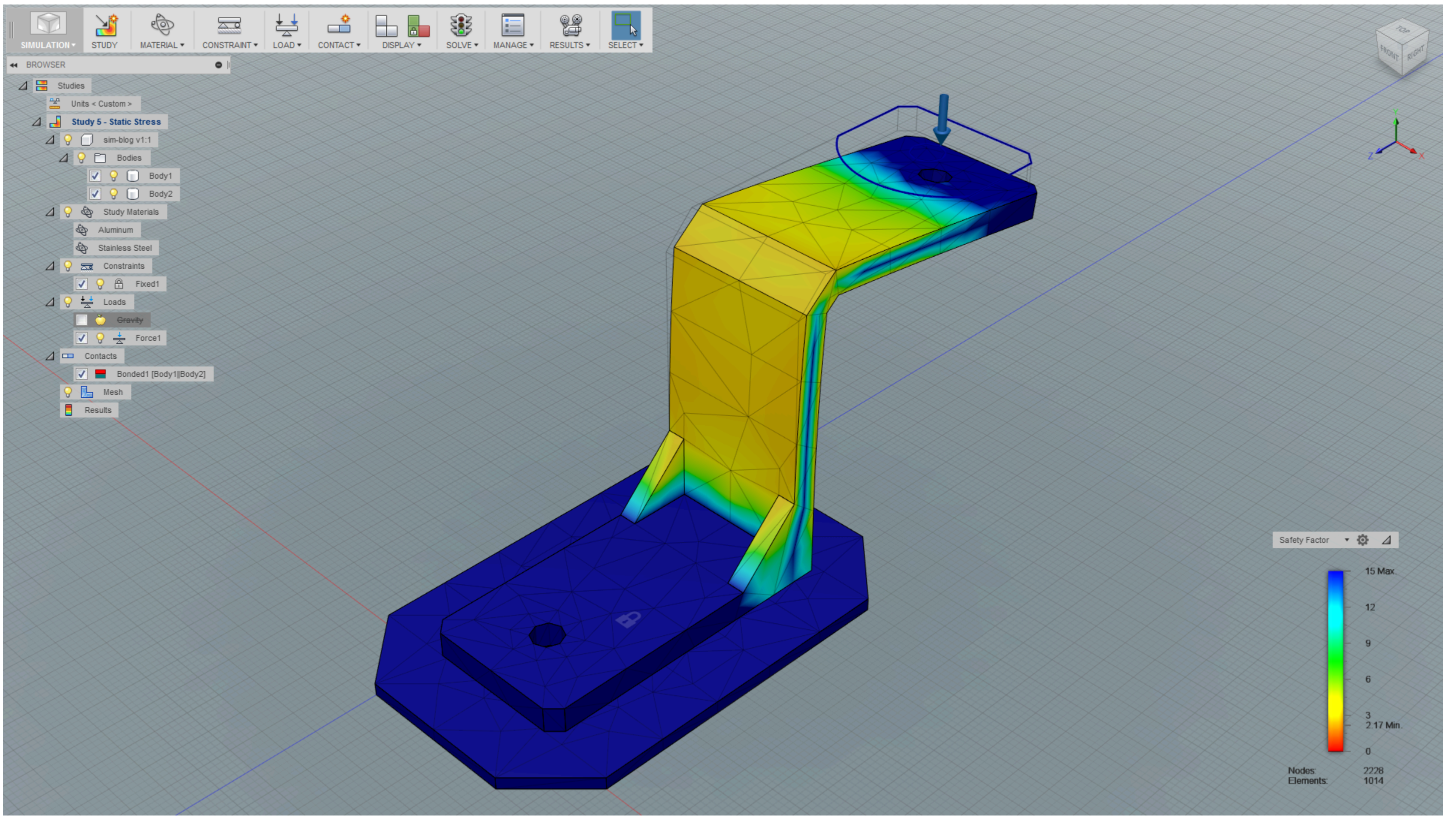
Proper constraints define how components interact. This section covers advanced contact sets, interference detection, and bill-of-materials generation.

✓ Contact sets & collision detection	✓ Flexible sub-assemblies
✓ Interference analysis workflow	✓ External references & linked designs
✓ Bill of Materials (BOM) generation	✓ Assembly section views

4.3 Motion Studies & Animation

✓ Joint limits & range of motion	✓ Exploded view creation
✓ Drive joints for animation	✓ Animation export for presentation
✓ Motion study timeline & keyframes	✓ Assembly drawing documentation

Module Duration: 10 Hours | Sessions: 5 | Lab Exercises: 4



Module 5 Computer-Aided Manufacturing (CAM)

Fusion's integrated CAM workspace bridges design and physical production. Learners program CNC toolpaths, simulate machining operations, and generate G-code for real-world manufacturing — all within the same design environment.

5.1 CAM Workspace & Setup

✓ CAM workspace introduction	✓ Stock definition & material setup
✓ Machining setup & WCS definition	✓ Machine configuration & post-processors
✓ Tool library management	✓ Feeds, speeds & cut parameters

5.2 2D & 2.5D Machining Operations

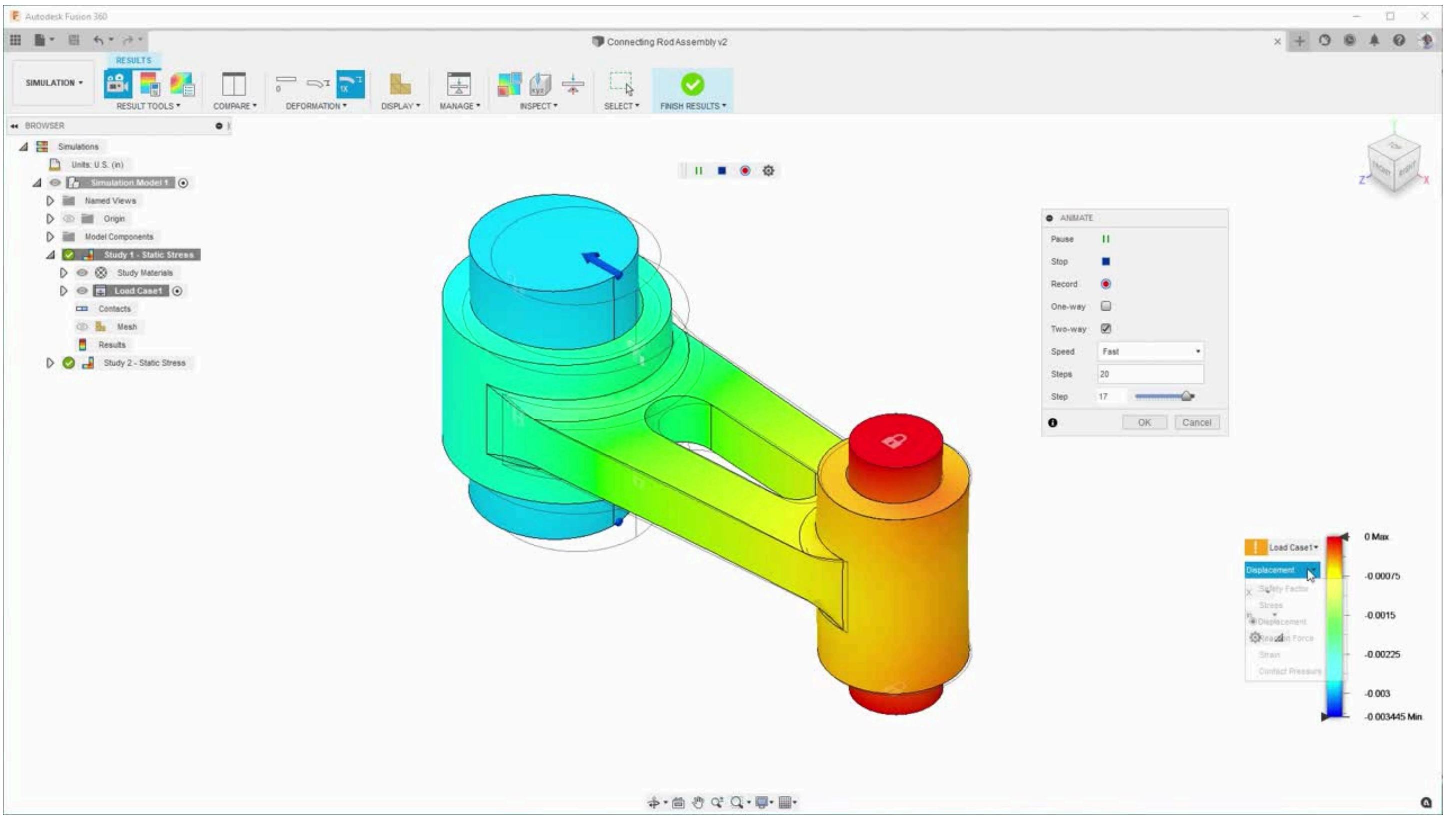
2.5D operations form the foundation of CNC machining. Participants learn to create facing, contouring, pocketing, and drilling operations for prismatic parts.

✓ Facing & adaptive clearing	✓ Thread milling operations
✓ 2D contour & pocket operations	✓ Engrave & trace toolpaths
✓ Drilling, boring & tapping cycles	✓ Rest machining strategies

5.3 3D Machining & Simulation

✓ 3D adaptive & parallel clearing	✓ Toolpath simulation & verification
✓ Scallop & pencil finishing passes	✓ G-code post-processing & export
✓ Horizontal & contour finishing	✓ DNC communication basics

Module Duration: 12 Hours | Sessions: 6 | Lab Exercises: 5



Module 6 Simulation & Structural Analysis (CAE)

Fusion's CAE simulation workspace enables engineers to validate design performance under real-world loading conditions before manufacturing. This module covers static stress analysis, modal analysis, thermal simulation, and shape optimization.

6.1 Simulation Fundamentals & FEA

✓ Introduction to Finite Element Analysis	✓ Mesh generation & refinement
✓ Simulation study types overview	✓ Boundary conditions & loads
✓ Material assignment & properties	✓ Fixed & pin constraints setup

6.2 Static Stress & Modal Analysis

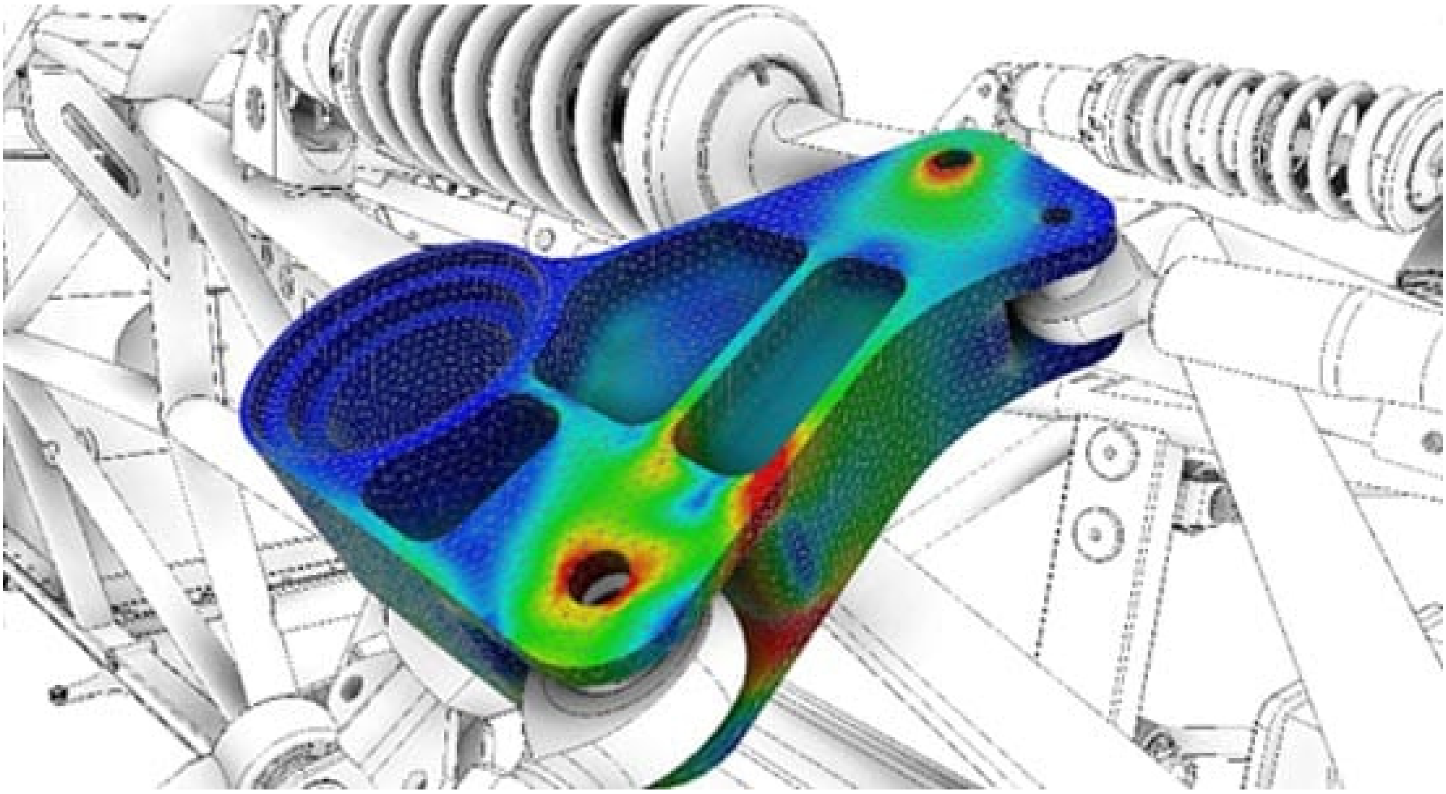
Static stress analysis evaluates structural integrity under applied loads. Modal analysis identifies natural frequencies to prevent resonance failures in dynamic applications.

✓ Linear static stress analysis	✓ Contact stress between bodies
✓ Factor of safety evaluation	✓ Modal frequency analysis
✓ Displacement & Von Mises stress	✓ Mode shape visualization

6.3 Thermal Analysis & Shape Optimization

✓ Steady-state thermal analysis	✓ Topology optimization workflow
✓ Heat flux & temperature loading	✓ Generative design overview
✓ Thermostructural coupling	✓ Results interpretation & reporting

Module Duration: 8 Hours | Sessions: 4 | Lab Exercises: 3



Module 7 Rendering & Product Visualization

Communicating design intent visually is a critical professional skill. This module covers Fusion's Render workspace to create photorealistic product visualizations, client presentations, and marketing materials directly from the 3D model.

7.1 Render Workspace & Materials

✓ Render workspace introduction	✓ Physically-based rendering (PBR)
✓ Appearance library & material assignment	✓ Environment & HDRI lighting
✓ Decals, textures & bump mapping	✓ Ray tracing & quality settings

7.2 Lighting, Camera & Scene Setup

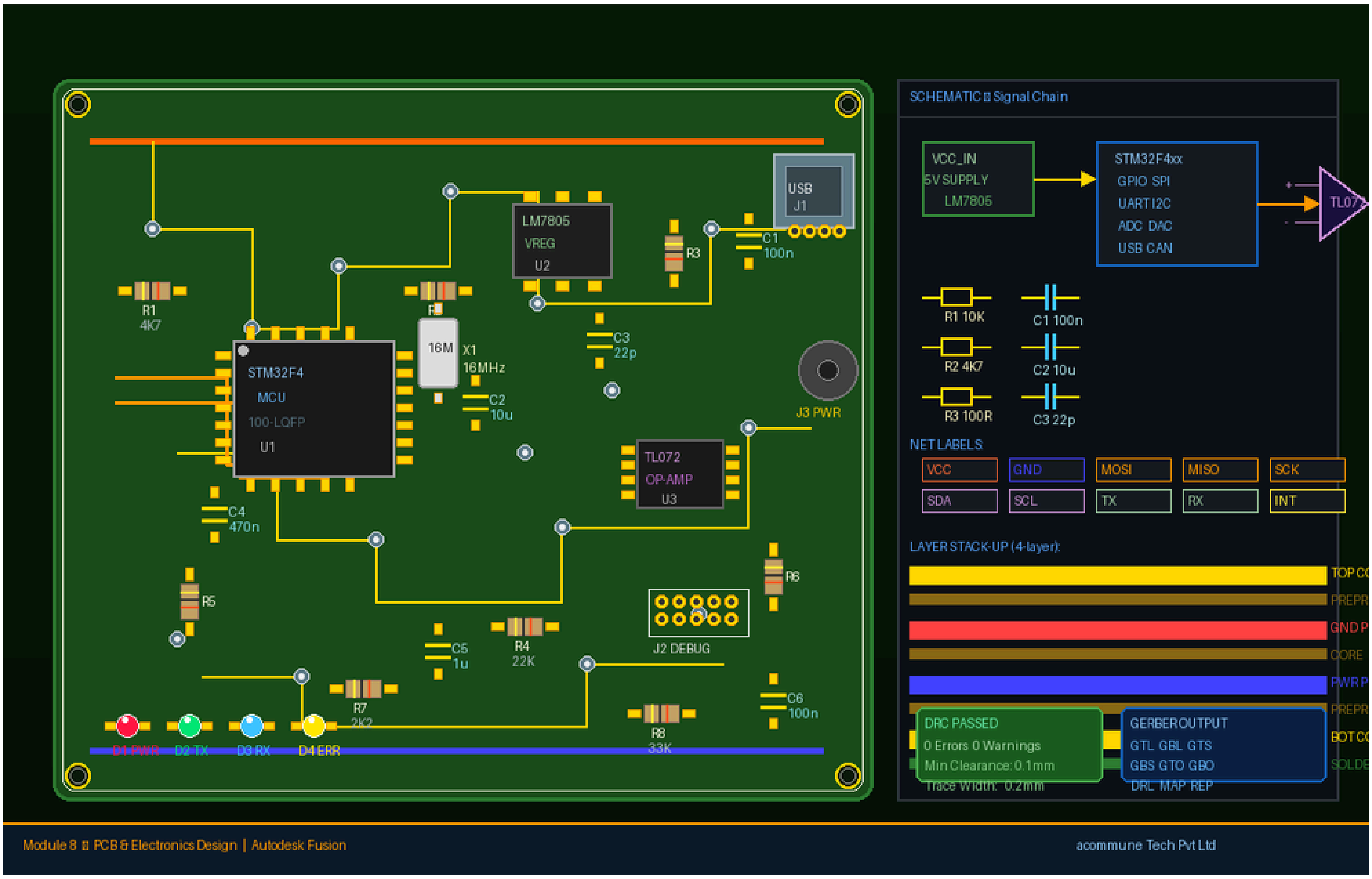
Professional renders depend on correct lighting and camera configuration. Learners set up photographic environments with studio-quality lighting and camera depth of field.

✓ Studio, outdoor & custom lighting	✓ Orthographic vs. perspective views
✓ Camera position & focal length	✓ Scene background customization
✓ Depth of field & exposure	✓ Turntable animation rendering

7.3 Cloud Rendering & Export

✓ Local vs. cloud rendering comparison	✓ Exporting stills & animations
✓ Autodesk cloud render credits	✓ Creating design presentations
✓ Render output settings & resolution	✓ Portfolio & client-ready output

Module Duration: 8 Hours | Sessions: 4 | Lab Exercises: 3



Module 8 PCB & Electronics Design

Autodesk Fusion uniquely integrates PCB layout and electronics design with mechanical 3D modeling, enabling concurrent electronics and enclosure design. This module introduces the Electronics workspace for schematic capture and PCB layout.

8.1 Electronics Workspace Fundamentals

✓ Electronics workspace overview	✓ Net connectivity & ERC checks
✓ Schematic editor introduction	✓ Schematic best practices
✓ Component library management	✓ 3D PCB integration concept

8.2 PCB Layout & Routing

PCB layout involves placing components and routing copper traces to connect them. Fusion's unified environment allows real-time PCB-to-enclosure fit checking.

✓ PCB board outline & layer setup	✓ Design Rule Check (DRC)
✓ Component placement strategies	✓ Copper pours & ground planes
✓ Manual & autorouting tools	✓ Gerber file generation & export

8.3 3D PCB & Enclosure Integration

✓ Pushing PCB layout to 3D model	✓ Clearance & interference checks
✓ ECAD-MCAD collaboration workflow	✓ PCB enclosure design workflow
✓ Component 3D body assignments	✓ Manufacturing output documentation

Module Duration: 10 Hours | Sessions: 5 | Lab Exercises: 4

Assessment Structure

All assessments are designed to evaluate both theoretical understanding and practical design competency. The final project requires learners to design, simulate, and manufacture-ready a real-world component from scratch.

✓ Module quizzes (8 x 10 marks)	✓ Final design project (100 marks)
✓ Practical lab evaluations (8 x 15 marks)	✓ Viva-voce & design review
✓ Mid-term project — Assembly (50 marks)	✓ Portfolio submission

Certification & Recognition

Upon successful completion of all modules and assessments with a minimum score of 60%, participants receive the acommune Tech Pvt Ltd Certificate of Completion in Autodesk Fusion. High performers (>85%) receive a Merit Certificate.

Certificate Name	Certificate in Autodesk Fusion — Professional Course

Issued By	acommune Tech Pvt Ltd, Training Division

Validity	Lifetime — No Expiry

Verification	Online verification portal: verify.acommune.tech

Contact	training@acommune.tech | +91-74492 76396

Prerequisites & Target Audience

✓ Basic computer literacy required	✓ Working professionals in manufacturing
✓ Engineering students (Mech, Prod, Design)	✓ Product designers & industrial designers
✓ No prior CAD experience needed	✓ Entrepreneurs & startup founders

Software & Tools Required

✓ Autodesk Fusion (latest version)	✓ Minimum 8 GB RAM (16 GB recommended)
✓ Autodesk account (free or paid)	✓ Dedicated GPU with 1 GB VRAM
✓ Windows 10/11 or macOS 12+	✓ Reliable internet connection

Learning Outcomes

Upon completion of this course, participants will be fully capable of:

1. Creating complex 3D parametric models using Autodesk Fusion's modeling tools
2. Building and animating multi-component assemblies with joints and motion studies
3. Programming CNC toolpaths and generating G-code for machining operations
4. Running structural FEA simulations and interpreting safety factors
5. Producing photorealistic renders for client presentations and portfolios
6. Designing schematic circuits and PCB layouts integrated with 3D enclosures

acommune Tech Pvt Ltd

www.acommune.tech | training@acommune.tech